

Weekly Update One

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Weekly Progress (5 points)

1. Solidifying a working list of variables necessary to collect for the initial dataset
2. Researching alternatives to gold price – also looking to pull the data from the Bloomberg terminal in Duques
3. Pulling the majority of variables for this dataset, taking some ideas from the some existing research found in the literature review of the project proposal – have a running list of potentials, but as of right now unnecessary
4. Filling in the gaps of daily data from FRED and Yahoo Finance as the weekends are empty – seeing if it is even necessary or simply redundant as rates and prices do not change on non-business days – Resample all monthly and weekly files
5. Determining the best way to join all of the individual data files together while ensuring accuracy – would it be easier to copy and past all the individual files and then merge with the file with solely fred and yahoo finance – still exploring
6. Joining all the data into one database

Challenges (3 points)

1. Resampling monthly data into daily data – In order to do so, I have to take each variable that I want to use and resample individually. I know I could probably make a function that takes all the individual csvs and resamples them immediately, however doing it manually ensures that there are less mistakes down the line when it comes to compiling into one large dataset
2. Figuring out the gold price issue – there are paywalls unfortunately with all financial data and I want to ensure accuracy
3. Underestimating the amount of time it would take me to collect and clean all of the data – in last week's elevator pitch, I thought my initial data collection and pre-processing would take much less time, but I did not budget as well as I wanted to with all of my other coursework (thus it was good that I allotted two weeks to do it all)
4. Figuring out exactly which variables and economic indicators would be worth my time collecting and cleaning – It becomes very difficult to discern as they all are technically important, but this is where feature selection comes in

For example, I know that the Factors Affecting Reserve Balances, known as the H.4.1 release from the Fed that occurs on a weekly basis is incredibly important for providing a snapshot of the Fed's balance sheet and its current holdings. Now, there are five major categories provided by FRED, but all of them have varying levels of importance. It is easier to just grab all of them and then resample them so they reflect the change for the weekly basis, but there is the question if it is redundant.

5. Thinking about the future of my project just in regards to the data and the resampling. I am thinking I use the same dataset for both an ARIMA and the MLM, however, MLMs can handle mixed frequency data whereas ARIMA cannot. I wonder how the accuracy would change with mixed frequency v same frequency data for the MLM.

Code Commits (2 points)

1. Data_Pulling_Resampling_Merging – the initial commits are combined into one.
2. Gold historical data

Other Potential Additions

For my sake I am going to list out all of the data I am currently thinking about adding, but not really certain if it is truly necessary.

Variables Maybe to Add

- Business investments – explore more options on what this could mean in terms of variables
- International Trade balance and Exchange value of U.S. dollar against major world currencies
- Manufacturing Trends

Other To-Do's

References